

Liam Llamazares-Elias

Curriculum Vitae

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Research Experience

Sep. 2025 – **Senior Research Associate**, *Department of Mathematics and Statistics*,
Aug. 2028 *Lancaster University*, Lancaster, UK
Postdoctoral position within the EPSRC Hub in Probabilistic AI (ProbAI Hub)
Supervised by Prof. Chris Nemeth, and Prof. Paul Fearnhead

Education

2021–2025 **PhD in Mathematical Modelling, Analysis and Computation**, *University of Edinburgh*, School of Mathematics
PhD with Integrated Study. Supervised by Prof. Finn Lindgren and Dr. Jonas Latz

2020–2021 **MA in Mathematical Modelling**, *University of Salamanca*
GPA: 9.85/10
Master's thesis: *Mathematical Theory in Non-linear Diffusion Processes, The Porous Medium*. Grade: 10/10 (Summa cum laude)
Link to masters thesis: [here](#)

2019–2020 **Master-level Courses (M1 and M2)**, *Département de Mathématiques*, France, Université de Poitiers
GPA: 17.97/20
Erasmus Scholarship

2016–2020 **BSc in Mathematics**, *University of Salamanca*
GPA: 9.14/10
Undergraduate thesis: *Some Results on the Existence and Uniqueness of Solutions to the Navier-Stokes Equations*. Grade: 10/10 (MH)
Link to undergraduate thesis: [here](#)

Research Visits

2023 **Participant in the Research Program**, *The Mathematical and Statistical Foundation of Future Data-Driven Engineering*, Isaac Newton Institute for Mathematical Sciences, University of Cambridge. June.

Publications

2026 **L. Llamazares-Elias**, J. Latz, F. Lindgren, *A Parameterization of Anisotropic Gaussian Fields with Penalized Complexity Priors*. Journal of the American Statistical Association. DOI; [arXiv](#).

2026 **L. Llamazares-Elias**, S. Llamazares-Elias, J. Latz, S. Klus, *Data-driven approximation of Koopman operators and generators: Convergence rates and error bounds*. Journal of Computational Dynamics, 15, 163–206. DOI; [arXiv](#).

- 2021 **L. Llamazares-Elias**, S. Llamazares-Elias, A. Martín del Rey, *An Analysis of Contact Tracing Protocol in an Over-Dispersed SEIQR Covid-Like Disease*, Physica A: Statistical Mechanics and its Applications. [DOI](#)

Scholarships and Awards

- 2021–2025 **MAC-MIGS PhD Scholarship**, University of Edinburgh, Maxwell Institute Graduate School
Competitive fully funded PhD position
- 2020–2021 **Collaboration Scholarship**, Department of Applied Mathematics, University of Salamanca
Awarded by the Spanish Ministry of Education (Only one scholarship assigned per department)
- 2020–2021 **Research Scholarship, Porous Medium Equation**, Institute of Theoretical Physics and Mathematics, University of Salamanca

Selected Talks and Presentations

- Jun 2026 Invited seminar: *Data-driven approximation of transfer operators*, AI Hub Seminar, Department of Statistics, University of Oxford.
- Apr 2026 *Non-stationary Gaussian Fields and Penalized Complexity Priors*, Computational Statistics and Machine Learning Reading Group, Lancaster University.
- Dec 2025 Invited talk: *Robots, simulación y optimización*, MUMOMA, University of Salamanca.
- 2025 *Data-driven approximation of Koopman operators and generators: Convergence rates and error bounds* at Hidden structures in dynamical systems, optimization, and machine learning workshop in L'Aquila. May.
- 2024 *Penalized Complexity Priors for Non-Stationary Gaussian Fields* at CMStatistics Conference in London. December.
- 2024 *Penalized Complexity Priors for Anisotropic and Non-Stationary Gaussian Fields* at Spatial Point Processes Reading Group. December.
- 2023 *A Parameterization of Anisotropic Gaussian Fields with Penalized Complexity Priors* at CMStatistics Conference in Berlin. December.
- 2023 *A Parameterization of Anisotropic Gaussian Fields with Penalized Complexity Priors* at Mathematics of Information & Data Science Seminar in Edinburgh. December.
- 2023 *A Parameterization of Anisotropic Gaussian Fields with Penalized Complexity Priors* at Numerical Analysis of Stochastic Partial Differential Equations (NASPDE) Workshop. April.

Teaching experience

- 2023–2025 Tutor of Statistical methodology at University of Edinburgh
- 2023–2024 Tutor of Applied Statistics at University of Edinburgh

Academic Service

- 2025–Present Organiser of the Computational Statistics and Machine Learning Reading Group, Lancaster University.
- 2026–2027 Working Group Member for the Prob_AI Hub Winter School in Bristol, contributing to the programme and speaker selection.
- 2022 Co-organiser of the Malliavin Calculus Reading Group at the University of Edinburgh, facilitating weekly discussions and presentations among PhD students. June–November.

Other Experience

- 2022–Present Author of mathematics blog on PDE and stochastic analysis. nowheredifferentiable.com
- 2023–2024 Participation in reading group on Mean field games and interacting particle systems. Held through the University of Edinburgh and Heriot–Watt University. October–February.

Other skills

- Languages English (Native), Spanish (Native), French (B2), Mandarin (B2 in reading and listening)
- Programming Experienced in use of Python, R, C, and Mathematica.